

Class XI | Unit 1 | weight 15%

Overview of the computer system

Board weight about 15% | 15 theory periods

A computer is a system, not a single box. This lecture names the parts, sorts the devices by the job they do, and separates the programs that run the machine from the programs that do your work.

What you should be able to do

- Describe a computer and the four basic operations.
- Draw the block diagram and place hardware, software and memory on it.
- Classify input, output, storage, processing and communication devices.
- Explain system software, application software and utility software.
- Compare primary and secondary memory.
- Describe how computing is used now, including artificial intelligence and cloud computing.

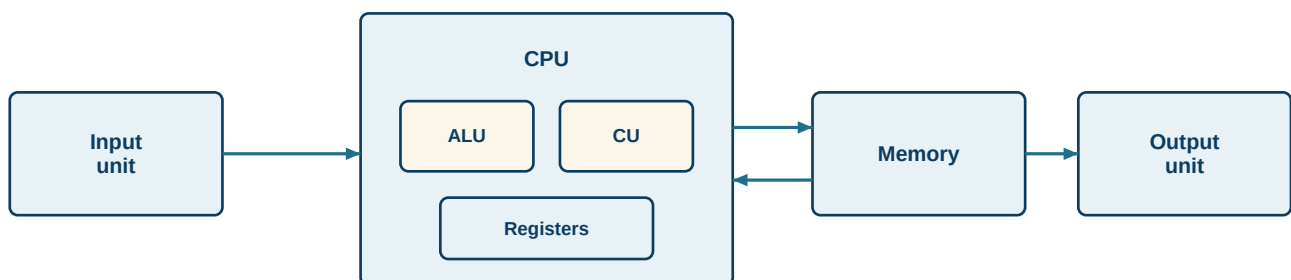
Path through the lecture

- 1 Start with the definition and the block diagram. Students copy the diagram before any list of devices.
- 2 Sort devices by function. A device can be named only after its class is clear.
- 3 Teach the three software families with one example each.
- 4 Close with artificial intelligence and the cloud, each tied to a local example.

1.1 A computer system

Computer. An electronic system that takes data as input, processes it according to instructions, stores data and results, and gives information as output.

Data is the raw fact you enter, such as a set of marks. Information is the processed result, such as a grade or a class average. The machine follows a program, which is a stored sequence of instructions. The four operations you should be able to name in order are input, processing, storage and output.



Data moves in. The CPU processes it with memory. Results go out.

Figure 1.1 Block diagram of a computer system.

The arrows in Figure 1.1 are the basic operations. Input devices send data in. The CPU processes it. Memory holds the instructions and the data while the work is going on. Output devices present the result. Secondary storage keeps a copy after the power is switched off. Software is the set of programs that tell this hardware what to do. Hardware is the physical equipment.

1.2 Hardware by function

Classify a device by the job it does. A keyboard is input because data travels from the user into the computer. A printer is output because information travels out. A hard disk is storage. The processor is the processing device. A router is a communication device.

Input devices

- **Keyboard** enters characters. A mouse points and clicks.
- **Joystick or joypad** is used for direction and actions, often in games and simulators.
- **Scanning devices** turn a printed page or a photograph into digital data.
- **Data tablet and stylus** let a person draw or write by hand. The stylus is the pen.
- **Voice recognition** turns speech into text or commands. A **microphone** captures the sound.
- **Touch screen** receives input by touch and often displays output as well. Name the role the question asks about.
- **Webcam** captures live video. A **barcode scanner** reads a printed code on a product.

Output devices

- **Monitors** display results. LED and LCD panels are flat and use less power. Plasma is an older flat panel. CRT is the deep glass monitor of earlier computers.
- **Printers** put results on paper. A dot-matrix printer is an impact printer: pins strike a ribbon. Inkjet and laser printers are non-impact. Thermal printers heat special paper.
- A **plotter** draws precise lines, used for maps and engineering drawings.
- A **multimedia projector** throws the screen onto a wall. **Speakers and headphones** produce sound.
- A **speech generating device (SGD)** speaks text aloud for a user who cannot use ordinary speech.

Storage, processing and communication

- **Hard disk** stores large amounts of data magnetically. **Flash, USB and pen drives** store data in electronic memory and travel in a pocket.
- **CD, DVD, CD-R and CD-RW** are optical discs. CD-R records once. CD-RW can be rewritten. **Blu-ray** holds high-definition video. **SD and microSD cards** are small flash cards used in phones and cameras.
- The **processor** is the processing hardware. It is taught properly in Unit 3.
- **Network interface card** connects the computer to a network cable or a wireless network.
- A **modem** turns the computer's digital signal into a form a telephone or cable line can carry, and turns the incoming signal back.
- **Data terminal equipment** is the device at the end of a link, such as a computer or a terminal. A **fax machine** sends a copy of a page over a telephone line.
- A **hub** repeats data to every connected device. A **switch** sends a frame only toward its destination inside a local network. A **router** forwards packets between networks. A **gateway** connects networks that do not speak the same protocol and translates between them.

Exam point

If a question asks you to classify webcam, plotter, SD card, router and keyboard, write the class first and the name second: input, output, storage, communication, input.

1.3 Software

Software. The programs and related data that tell the hardware what to do.

System software runs and controls the computer. It includes the operating system, device drivers, language translators and antimalware programs. A device driver teaches the operating system how to talk to one piece of hardware, such as a printer. A language translator turns a program written by a person into a form the processor can run.

Language translators

Translator	What it does
Assembler	Converts assembly language into machine code.
Compiler	Translates a whole high-level program into an object program, then the program can be run.
Interpreter	Translates and runs one statement at a time. It stops at the first error it meets.

Antimalware protects the computer from harmful programs. Antivirus software finds and removes viruses. A Trojan horse is a harmful program disguised as something useful, so protection has to recognise that disguise as well.

Application software does a job for the user. General-purpose software is sold for many people and many tasks: a word processor, a spreadsheet, a browser. Special-purpose software is built for one kind of organisation: a college admission system, a hospital record system, a shop billing system.

Utility software maintains the computer. Compression software makes files smaller for storage or sending. Data-recovery software tries to bring back deleted or damaged files. A disk manager partitions a disk, cleans up files and can defragment a hard disk.

1.4 Primary and secondary memory

Primary memory is the memory the CPU works with directly. RAM holds the running program and its data. It is fast, and it is volatile: the contents go when the power goes. ROM holds permanent start-up instructions and is non-volatile.

Secondary memory keeps data and programs for the long term. Hard disks, USB drives, optical discs and memory cards are secondary. The CPU does not execute a program from a USB drive directly. The program is first loaded into RAM. Secondary memory has more capacity, costs less per byte, and keeps its contents without power. It is slower than RAM.

Primary memory and secondary memory

Basis	Primary	Secondary
Contact with the CPU	Direct	Indirect, through primary memory
Speed	Faster	Slower
Typical contents	Running program and data	Stored files and software
Power removed	RAM loses its contents	Contents remain
Capacity and cost	Smaller, costlier per byte	Larger, cheaper per byte

1.5 Computing now

Computers now sit inside phones, vehicles, hospitals, banks and classrooms. They speed up calculation, store huge records, and connect people. The same tools raise questions of privacy, honest use of information, and dependence on a working network.

Artificial intelligence. The building of computer systems that perform tasks which ordinarily need human intelligence, such as recognising speech, learning from examples, and supporting a decision.

In a hospital, an AI system can flag an X-ray for a doctor to review. In a school it can practise language with a student. In traffic it can change signal timing. On a farm it can watch crop images for disease. The system is a tool for a person who remains responsible for the decision.

Cloud computing. The delivery of storage, software and processing over a network, from servers run by a provider, paid for and used as they are needed.

- A college can start with a small service and increase storage or processing when admissions begin.
- Students reach the same files from a lab computer or a phone.
- The provider maintains the servers and keeps backup copies as part of the service.
- Several people can work on one document.
- The college spends less on its own server room at the start.

Cloud work needs a network connection. The data lives on the provider's machines, so the college needs a clear agreement about who may see it.

Exam point

A strong 8-mark answer on this unit is the block diagram with labels, plus one paragraph that places software and both kinds of memory on the diagram.

Key terms

Term	Meaning to write in an answer
Data	Raw facts entered into the computer.
Information	Data after processing, useful for a decision.
Hardware	Physical parts of the computer system.
Software	Programs that instruct the hardware.
System software	Programs that operate and control the computer.
Application software	Programs that perform a user task.
Utility software	Programs that maintain the computer.
Volatile memory	Loses its contents when power is removed.
Cloud computing	On-demand computing services delivered over a network.

Bring it together

- A computer inputs, processes, stores and outputs. Hardware does the physical work. Software gives the instructions.
- Name every device by its function: input, output, storage, processing or communication.
- System software runs the machine. Application software does the user's job. Utilities maintain the machine.
- Primary memory serves the CPU now. Secondary memory keeps files for later.
- AI systems support intelligent tasks. Cloud services provide computing over a network.

Do this in the practical

- 1 On a real system unit, point to the processor area, the memory modules, the hard drive and one port. Say which block of Figure 1.1 each part belongs to.

Check yourself

Attempt every item before you read the key. Section A of the Computer Science paper is a set of multiple-choice questions.

1. Which operation turns data into information?

- A. Input
- B. Processing
- C. Storage only
- D. Printing only

2. A plotter is classified as:

- A. Input
- B. Output
- C. Processing
- D. Communication

3. Which program translates a high-level program one statement at a time?

- A. Compiler
- B. Assembler
- C. Interpreter
- D. Device driver

4. A college fee system written for that college only is:

- A. Special-purpose application software
- B. An operating system
- C. A device driver
- D. ROM

5. RAM is described as volatile because:

- A. It is very large
- B. Its contents are lost when power is removed
- C. It stores files permanently
- D. It cannot be read

6. Which device forwards packets from one network to another?

- A. Plotter
- B. Router
- C. Stylus
- D. Compiler

7. Cloud computing provides resources:

- A. Only by replacing the keyboard
- B. Over a network, as they are needed
- C. Only on a CD-ROM
- D. Only inside ROM

8. Compression software is a:

- A. Utility
- B. Language translator
- C. Input device
- D. Protocol

Answer key

Q	Answer	Why this is the answer
1	B	Processing applies instructions to data and produces information.
2	B	A plotter presents results, usually as a precise drawing.
3	C	An interpreter translates and executes statement by statement.
4	A	It performs one organisation's task, so it is special-purpose application software.
5	B	Volatile memory needs power to hold its contents.
6	B	A router connects networks and forwards packets.
7	B	The provider delivers storage, software or processing over a network.
8	A	Compression maintains files by making them smaller. That is a utility.

Short answers

A 3-mark answer needs a definition and two clear points, or a three-row difference. Write in sentences.

Q1. Define a computer.

Model answer

A computer is an electronic system that accepts data, processes it according to stored instructions, stores data and results, and produces information as output.

Q2. Differentiate compiler and interpreter.

Model answer

A compiler translates the whole source program and produces an object program which is then executed. An interpreter translates and executes one statement at a time and reports an error as soon as it reaches it. A compiled program is generally ready to run again without translating the source each time.

Q3. Give two differences between primary and secondary memory.

Model answer

Primary memory is directly accessible by the CPU and is faster. RAM, which is primary memory, loses its contents when power is removed. Secondary memory keeps files after power is removed, has larger capacity, and is slower. Programs are loaded from secondary memory into primary memory before execution.

Q4. What is a device driver?

Model answer

A device driver is system software that lets the operating system control a particular hardware device, such as a printer or a network card.

Q5. Write two advantages of cloud computing.

Model answer

The user can increase storage or processing when the work grows, and can reach the same files from different places. The provider maintains the servers, so the user spends less on a private machine room at the start.

Detailed answers

Section C asks for a connected answer. Open with a definition, develop the points below, and close with one example.

Question 1. Draw the block diagram of a computer system and explain its parts.

- 1 Define a computer in one sentence.
- 2 Draw input, CPU, memory and output. Inside the CPU show ALU, control unit and registers. Show a double arrow between CPU and memory.
- 3 Explain input as data entering, processing as the CPU following instructions, memory as the working store, and output as information leaving.
- 4 Add secondary storage under memory and say that it keeps files when power is off.
- 5 Close with one line: hardware is the equipment in the diagram, software is the instructions that make it work.

Question 2. Explain system software, application software and utility software with examples.

- 1 Define software.
- 2 System software operates the computer: operating system, device drivers, translators (assembler, compiler, interpreter) and antivirus.
- 3 Application software does user work. General-purpose examples include a word processor and a browser. A special-purpose example is a hospital system.
- 4 Utility software maintains the machine: compression, data recovery and disk management.
- 5 Give one situation in which a student uses one program from each family on the same day.